


**Team Project 1**

Team 15 - Gigi Winter, Gaby Thorn, Ryan Loev, Max Fisher

MIS 38200-002 - Fall 2026

October 18, 2026

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**Client Research & Value Chain Analysis (10 pts)**

Eli Lilly and Company is a global pharmaceutical company headquartered in Indianapolis, Indiana. Founded in 1876, the company focuses on researching, developing, manufacturing, and distributing medications for a variety of medical conditions. Some of Lilly's major areas of focus include diabetes, obesity, cancer, and immunological diseases. The company operates in the pharmaceutical industry, where product development, manufacturing quality, and regulatory compliance are all important to making sure medications are delivered safely and effectively.

Two of Lilly's most significant products are Mounjaro and Zepbound, which both contain the active ingredient trizepatide. Mounjaro is primarily used to treat type 2 diabetes, while Zepbound is used for weight management and certain other approved conditions. Demand for these medications has grown rapidly, creating challenges for Lilly as it works to increase manufacturing capacity and keep its products available to patients.

Unlike many other industries, pharmaceutical companies cannot quickly increase production when demand rises. Manufacturing medications requires specialized equipment, strict quality standards, regulatory approval, and significant investments in facilities. New manufacturing plants can take years to construct and prepare for commercial production. Because of this, unexpected increases in demand can create shortages that are difficult to resolve immediately.

Lilly has experienced these challenges firsthand. In April 2024, shortages of certain doses of Zepbound were reported as demand continued to grow. The company has since made substantial investments in expanding its manufacturing network to support current and future demand. Although the FDA determined that the national tirzepatide shortage was resolved in December 2024, the situation showed how difficult it can be to keep production aligned with rapidly changing customer demand.

Lilly's operations involve several interconnected departments, including research and development, supply chain management, manufacturing, quality assurance, inventory management, and distribution. Each department plays an important role in the process of getting medications from development to patients. For this project, our team chose to focus on manufacturing, specifically the production planning process for Mounjaro and Zepbound. We identified production planning as an area where AI could be useful because planners must consider several factors, including demand forecasts, inventory availability, production schedules, and possible disruptions, before deciding how to respond to changes in supply and demand.

One of Lilly's major strategic priorities is expanding its manufacturing capabilities to meet growing demand. Since 2020, the company has announced more than $55 billion in manufacturing investment commitments worldwide. These investments include new facilities, expanded production capacity, and advanced manufacturing technologies. The goal is to increase

the amount of medicine Lilly can produce while continuing to meet the quality and safety standards required in the pharmaceutical industry.

Lilly's strategy also focuses on improving patient access to medications. When demand exceeds available supply, patients may have difficulty obtaining their prescribed treatments. This makes production planning especially important because the company must determine how to use its available manufacturing resources to meet demand as effectively as possible.

While building additional facilities will help Lilly increase production in the long term, these investments do not immediately solve short term capacity constraints. In the meantime, effective production planning can help the company make better use of its existing resources. Our proposed AI application supports this strategy by helping planners evaluate possible inventory shortages, changes in customer demand, and production disruptions before making decisions. Rather than automatically changing production schedules, the application provides information and recommendations that planners can review when deciding what actions to take.

Lilly's manufacturing value chain consists of several stages that work together to make sure medications are produced and delivered to customers. For this project, we identified eight major stages: Demand Forecasting → Production Planning → Raw Material Supply → Manufacturing → Quality Assurance → Inventory and Storage → Distribution → Customer Demand. Each stage plays an important role in meeting customer demand. However, there are also opportunities for inefficiencies involving time consumption, human error, resource allocation, and technology.

Our team chose to focus on the connection between demand forecasting, production planning, and inventory management. These stages depend heavily on accurate information and making decisions in a timely manner. If customer demand increases unexpectedly or manufacturing output decreases, production planners must determine whether existing inventory and scheduled production will be enough to meet demand. They also need to consider whether the remaining inventory will meet safety stock requirements. Without an efficient way to evaluate these factors together, planners may spend additional time reviewing data, performing calculations, and comparing different options. There is also the possibility of human error when calculating inventory levels or making assumptions about future demand, which could affect production decisions.

To address these potential inefficiencies, our team developed an AI production planning application using Dify.ai. Our consulting team, Karate Jones, designed the application to help users evaluate production planning scenarios for Mounjaro and Zepbound. The application combines simulated manufacturing data, inventory calculations, knowledge retrieval, and AI generated recommendations into one workflow. The purpose is to help users identify potential production risks and better understand what actions could be taken before making decisions.

The application begins by asking the user to enter a business question, select a product, choose a planning week, and provide two scenario inputs. These inputs are the percentage change in demand and the percentage of planned production that is disrupted. This allows users to test different situations instead of relying on one fixed demand forecast. For example, a user could see how a 15% increase in demand or a 10% production disruption would affect projected inventory for a particular week.

After receiving the inputs, the application moves through a knowledge retrieval node containing research documents related to Eli Lilly, its products, and other relevant business information. This provides a source of research that could be used to support the explanations and recommendations generated by the AI assistant. The workflow then moves into a Python code node, which performs the inventory calculations using simulated planning data. By completing these calculations through code rather than asking the AI model to calculate everything itself, the application can produce more consistent numerical results.

Because Lilly's actual manufacturing schedules, inventory records, and production capacities are not publicly available to our team, we created a simulated dataset for the application. The dataset includes four planning weeks for both Mounjaro and Zepbound. Each product and week combination has an assigned starting inventory, forecasted demand, planned production, and safety stock target. These values are hypothetical and are meant to represent the types of information manufacturing planners would normally use when evaluating production schedules and inventory availability.

For example, in the first planning week, our simulated data assumes that Mounjaro has a starting inventory of 20,000 units, forecasted demand of 100,000 units, planned production of 90,000 units, and a safety stock requirement of 10,000 units. With no changes in demand or production, the projected ending inventory would be 10,000 units. This meets the simulated safety stock requirement and would be classified as low risk.

For Zepbound, the first planning week assumes starting inventory of 30,000 units, forecasted demand of 180,000 units, planned production of 120,000 units, and safety stock of 15,000 units. Under these conditions, projected ending inventory would be negative 30,000 units, meaning the modeled supply would not be enough to meet the forecasted demand. The application would classify this scenario as high risk because it indicates a possible shortage of 30,000 units.

The application uses several calculations to evaluate each scenario:

Adjusted Demand = Forecasted Demand × (1 + Demand Change % / 100)

Adjusted Production = Planned Production × (1 − Production Disruption % / 100)

Projected Inventory = Starting Inventory + Adjusted Production − Adjusted Demand

Inventory Cushion = Projected Inventory − Safety Stock

These calculations allow the application to consider both changes in customer demand and reductions in planned production. The inventory cushion also helps determine whether the projected ending inventory will meet the desired safety stock level.

Once the calculations are completed, the workflow uses an IF/ELSE decision node to classify the scenario into one of three risk levels. A high risk scenario occurs when projected inventory is negative, meaning there is not enough modeled supply to meet forecasted demand. A medium risk scenario occurs when projected inventory is zero or greater but remains below the required safety stock level. A low risk scenario occurs when projected inventory meets or exceeds the safety stock requirement. These risk classifications were created by our team for the project and do not represent Eli Lilly's actual internal risk policies.

Depending on the calculated risk level, the workflow directs the scenario to one of three separate AI analysis nodes. Each node uses a GPT model with instructions designed for its specific risk category. High risk scenarios focus on identifying possible shortages and determining what actions could help reduce the impact. Medium risk scenarios focus on protecting safety stock levels and preventing inventory conditions from getting worse. Low risk scenarios focus on maintaining stability, monitoring demand, and avoiding unnecessary increases in production.

Our AI assistant, Karate Jones, then takes the calculated results and provides an explanation of the scenario. The response includes a mission brief, situation analysis, explanation of why the results matter, three recommended actions ranked by priority, and the next decision that a planner should consider. Each recommendation explains the potential benefits, possible tradeoffs, and additional information that may be needed before taking action. The goal is to make the results easier to understand and give planners a starting point for evaluating different options.

The application is meant to support decision making rather than automatically make changes to production. For example, if a scenario indicates high risk, the assistant might recommend reviewing production schedules, checking demand forecasts, or determining whether additional production is possible. However, the application does not assume that manufacturing capacity can be increased immediately or that production changes can happen without considering pharmaceutical quality and regulatory requirements. These decisions would still need to be reviewed and approved by qualified employees.

The initial model evaluates one planning week at a time and assumes that starting inventory is available for use and planned production can be completed unless it is reduced by the user's disruption percentage. It does not currently account for every real world manufacturing constraint, such as equipment compatibility, supplier availability, production lead times, product release delays, or detailed manufacturing capacity limits. These factors would need to be included before the application could be used for actual production planning.

Overall, our Dify.ai application shows how AI could help improve the production planning portion of Lilly's value chain by combining inventory calculations with business recommendations. The application gives users a consistent way to identify potential inventory risks, reduces the need to manually perform repeated calculations, and makes it easier to understand how changes in demand or production could affect available inventory. Although the current version uses simulated data, it demonstrates how an AI application could help planners save time, reduce the chance of overlooking inventory problems, and make more informed decisions when production conditions change.

**Problem Identification & Strategic Alignment (10 pts)**

• Problem is clearly articulated with compelling explanation of its impact
• Solution aligns explicitly and persuasively with the client's strategy

The main problem our team identified is that Lilly's production planning process can struggle to keep manufacturing output in line with changing demand for Mounjaro and Zepbound. Production planning sits between demand forecasting and manufacturing in the value chain, so it's the step where forecasts get turned into actual decisions about what to produce, how much to make, and when. If this step can't react quickly or accurately when demand shifts, the problem carries into every stage after it.

Lilly's recent history shows how costly this gap can be. Mounjaro first appeared on the FDA's drug shortage list in 2022, and by April 2024, most doses of both drugs were in shortage. This happened only a few months after Zepbound was approved, at a time when the company had said it was confident supply would be fine. The FDA kept pushing back its estimated end date for the shortage as demand continued to outpace supply. Lilly itself reported that uptake of tirzepatide was about four times higher than its previous launches, which shows how far actual demand ended up from what the company had planned for.

The impact of these shortages reached several groups:

- **Patients** had to search for pharmacies that still had the drug in stock, and some doctors had to switch patients to different medications, which interrupted their treatment.
- **Lilly** lost potential revenue, and the shortage gave compounding pharmacies an opening to sell cheaper versions of tirzepatide that competed directly with Lilly's own products.
- **Planning accuracy** suffered in both directions. After the shortage ended, Lilly's sales came in lower than expected because wholesalers were selling through the inventory they already had instead of ordering more. Neither Lilly nor analysts saw this coming, which shows the problem isn't just producing too little. It's also about not having a clear picture of where supply actually sits in the market.

**Solution Description**

Our app is called Karate Jones. It's a Dify workflow built to help a production planner figure out whether Mounjaro or Zepbound is going to run short in a certain week, and what they could do about it.

The planner starts by picking a product, choosing a planning week from W1 to W4, and typing a question about the situation they want to look at. They also enter two percentages. First, they enter a predicted or measured demand change so they can see what happens as demand changes. Secondly, they enter a production disruption, which lowers output to represent things like equipment downtime or other delays.

Once the inputs are in, the app looks through the research our team uploaded on Lilly and the tirzepatide shortage, so the AI has some real background when it writes its response. Then the AI uses whatever parameters we give it; That includes how much inventory is on hand at the start of the week, how much demand is expected, how much is planned to be produced, and how much safety stock we want to keep. It raises or lowers the demand and production numbers based on the percentages the planner typed in. Then it figures out how much inventory would be left at the end of the week by adding production to the starting inventory and subtracting demand. Last, it checks how that leftover amount compares to the safety stock target.

The app uses that number to give the week a risk level. If projected inventory drops below zero, it's high risk, since that means a likely shortfall. If inventory stays positive but ends up under the safety stock target, it's medium risk. If it meets or beats safety stock, it's low risk. Each risk level goes to its own AI step with instructions written for that situation. For example, the high-risk version focuses on recovery options, and the low-risk version is told not to push for extra production when it isn't needed.

The final output is a short report written by "Karate Jones," a consultant character we gave a light Indiana Jones and Karate Kid theme so the results are a little more memorable. Every report has the same sections: a mission brief, a breakdown of the numbers, why the result matters, three ranked recommendations with their trade-offs, and the next decision the planner needs to make. It ends with a note that the data is simulated. We also told the AI to only use the calculated numbers, not to make up any figures about Lilly, and not to recommend anything that would skip quality or regulatory steps. Every recommendation is framed as an option for a person to decide on.

The biggest strength is that the math is done in code instead of by the AI. The numbers come out the same every time, and the AI is only there to explain them, so there's no risk of it miscalculating. The scenario inputs are also useful because a planner can test a demand jump and a production problem at the same time. That's close to what actually happened with tirzepatide, when demand took off while capacity couldn't keep up. We also liked that the app looks at safety

stock and not only outright shortages. A week where inventory technically stays positive but the buffer is almost gone still gets flagged, which gives planners time to react. Finally, because each risk level has its own instructions, the advice matches the situation instead of being the same generic response every time.

The app also has real limitations. All of the data is made up by our team, since Lilly doesn't make its inventory or production numbers public, so we haven't been able to test it on real figures. It only covers two products over four weeks, and that data is written directly into the code. A version Lilly could actually use would need to pull from their planning systems. The model is also pretty simple. It looks at one week at a time and doesn't carry leftover inventory into the next week. It also ignores things like dose strengths, which lines can make which product, raw material supply, and how long quality release takes. Production disruption is just one percentage, so the app can't tell the difference between an equipment failure and a supplier delay. The high, medium, and low cutoffs are our own assumptions, not Lilly's. And even with the guardrails, the AI could still suggest something that wouldn't work in a real plant, so a planner would need to check its recommendations before acting on them.

• **Working, published link:** Dify.AI Link - Karate Jones Consulting

• **All files necessary for testing are provided:**

- https://docs.google.com/document/d/1mYgMo-uWOVD_vlEjVkqD7GBut982fs-w_qauSGFAu-E/edit?usp=drivesdk
- https://docs.google.com/document/d/1CfUZ3XN4rgBfDJ4S9ipB6ZpjVtdnbTD7QhNeUAI3ywk/edit?usp=drivesdk
- https://docs.google.com/document/d/1MMKvddO2y9qCGaI0jKbGujExvr8iLYkRzvhlhPQI6rA/edit?usp=drivesdk

| Product | Week | Demand Change | Disruption | Projected Inventory | Safety Stock | Expected Risk |
|---|---|---|---|---|---|---|
| Zepbound | W1 | 0% | 0% | −30,000 | 15,000 | High |
| Mounjaro | W3 | 0% | 0% | 3,000 | 12,000 | Medium |
| Mounjaro | W2 | 0% | 0% | 15,000 | 10,000 | Low |
| Mounjaro | W1 | +10% | 0% | 0 | 10,000 | Medium |
| Mounjaro | W2 | 0% | 20% | −4,000 | 10,000 | High |

*Table 1. Product Expected Risk Levels by Week*

**Success Factor Rubric and Scale-Up Plan (10 pts)**

To determine whether our Dify.ai application is successful, our team developed several success factors that focus on improving the production planning process for Eli Lilly. The main goal of Karate Jones is to help production planners evaluate potential inventory shortages, understand changes in demand and production, and make more informed decisions. Since our application is currently using simulated data, we will measure its performance based on how accurately it calculates inventory levels, identifies risk, provides recommendations, and reduces the amount of time needed to analyze different scenarios.

We identified five key performance indicators (KPIs) that can be used to evaluate the success of our application. These include calculation accuracy, risk classification accuracy, time savings, recommendation quality, and system reliability. Each factor is directly related to the production planning challenges that our team identified earlier in the project.

1. Calculation Accuracy

One of the most important success factors is whether the application correctly calculates adjusted demand, adjusted production, projected inventory, and inventory cushion. Since these calculations determine the risk level assigned to each scenario, any errors could lead to incorrect recommendations.

To measure calculation accuracy, our team will create a set of test scenarios using our simulated data and calculate the expected results separately in Excel. We will then compare those results to the outputs generated by the Python code node in Dify.ai. Our goal is to achieve 100% calculation accuracy across all valid test scenarios. This is especially important because the application is intended to help planners make decisions based on numerical information.

2. Risk Classification Accuracy

Another important success factor is whether the application correctly identifies scenarios as high, medium, or low risk. These classifications determine which AI analysis node the scenario is sent to and what type of recommendations the user receives.

We will test this by entering different demand changes and production disruptions that produce each of the three risk levels. We will also test scenarios where projected inventory is exactly zero or equal to the safety stock requirement to make sure the classifications follow our established rules. Our goal is for 100% of valid test scenarios to be assigned to the correct risk category.

3. Time Savings

One of the main reasons our team chose production planning as an opportunity for AI is because reviewing inventory information and comparing different scenarios can take time. Our

application combines calculations, risk identification, and recommendations into one workflow, which could reduce the amount of manual work required.

To measure time savings, we will compare how long it takes a user to complete a production planning analysis manually in Excel versus using Karate Jones. Both methods will use the same simulated data and scenarios. We will record the time needed to complete each analysis, including calculating projected inventory, determining the risk level, and identifying possible actions. Our initial goal is to reduce the average analysis time by at least 30%. This target is an assumption created by our team and would need to be tested before claiming that the application actually saves time.

4. Recommendation Quality

Although the Python code performs the calculations, the AI assistant is responsible for explaining the results and recommending possible actions. Therefore, the quality of these recommendations is another important measure of success. The recommendations should be relevant to the risk level, easy to understand, and realistic given the information available.

To measure this, our team will review the AI responses using a five point scoring system. Each response will be evaluated based on its accuracy, relevance, clarity, practicality, and acknowledgment of limitations. For example, a high risk scenario should include recommendations related to possible inventory shortages, while a low risk scenario should focus more on monitoring and maintaining stable operations. We will also check that the AI does not invent manufacturing capacity, supplier information, or actual Eli Lilly inventory figures. Our goal is for the application to receive an average score of at least 4 out of 5 across the test scenarios.

5. System Reliability

The final success factor is whether the application consistently completes the workflow without errors. Users should be able to enter a valid scenario and receive a complete response without having to restart the application or manually correct an issue.

We will measure system reliability by running multiple test scenarios across different products, planning weeks, and risk levels. We will record how many tests successfully complete the entire workflow, from user input to the final AI response. Our goal is for at least 95% of valid test runs to complete successfully. We will also record any errors to identify whether they are caused by input validation, the Python code, knowledge retrieval, or the AI analysis nodes.

| Success Factor | Measurement Method | Target |
|---|---|---|
| Calculation Accuracy | Compare Dify calculations to independently verified Excel results | 100% accurate |
| Risk Classification Accuracy | Compare assigned risk levels to expected classification | 100% correct |
| Time Savings | Compare average completion time using Excel and Dify | At least 30% reduction |
| Recommendation Quality | Evaluate responses using a five point scoring rubric | Average of 4 out of 5 |
| System Reliability | Track successful workflow completions across valid test runs | At least 95% success |

*Table 2. System Reliability KPI's*

These KPIs will allow our team to evaluate the application using measurable results rather than simply determining whether the AI produces a response. We will use the results to identify which parts of the workflow need improvement before considering a larger implementation.

Although our current application demonstrates how AI could assist with production planning, it is still a prototype that uses simulated data for Mounjaro and Zepbound. Before an application like Karate Jones could be used in an actual manufacturing environment, several improvements would be needed. Our scale-up plan focuses on gradually moving from a classroom demonstration to a more realistic production planning tool while making sure the information is accurate, secure, and useful to employees.

The first step would be to improve the data used by the application. Currently, our Python code contains a fixed dataset with four planning weeks for each product. This is useful for testing different scenarios, but it does not reflect the constant changes that occur in a real manufacturing environment. For a larger implementation, the application would need access to updated inventory records, demand forecasts, production schedules, and safety stock requirements. Ideally, this information would come from an approved company database or enterprise resource planning (ERP) system rather than being manually entered into the code.Another important improvement would be connecting the knowledge retrieval system more directly to the AI analysis. Our current workflow includes research documents related to Eli Lilly, but the retrieved information is not yet passed directly into the final AI analysis nodes. As the application develops, we would want the AI to use approved documents, planning guidelines, and other relevant information when generating recommendations. This could make the responses more

useful because the assistant would be able to reference actual planning constraints and company procedures instead of relying only on the calculated inventory results. The application would also need to account for additional manufacturing constraints. Right now, the model considers starting inventory, forecasted demand, planned production, safety stock, demand changes, and production disruptions. However, actual production planning involves many other factors, including manufacturing capacity, equipment availability, supplier lead times, production schedules, and quality requirements. Adding these constraints would allow the application to evaluate more realistic situations and provide recommendations that better reflect what a manufacturing facility can actually accomplish.Security would also be an important consideration. Pharmaceutical companies work with sensitive operational information, and inventory records, production schedules, and manufacturing plans should not be accessible to unauthorized users. Before using actual company data, Lilly would need to evaluate the security of the application, determine what information the AI model is allowed to access, and establish appropriate access controls. The company would also need to review the application's data handling procedures and make sure they meet internal cybersecurity and compliance requirements.Another part of scaling up would involve testing the application with actual production planners. Our team can test whether the calculations and risk classifications work correctly, but employees who understand the manufacturing process would be needed to evaluate whether the recommendations are realistic. Their feedback could help identify missing information, improve the AI prompts, and make the application easier to use. The goal would be to develop a tool that supports the way planners already work rather than requiring them to completely change their existing processes.

| Phase | Timeline | Key Activities and Milestones |
|---|---|---|
| Phase 1: Prototype Testing | Month 1 | Test calculations, risk classifications, AI responses, and workflow reliability using simulated data |
| Phase 2: Data Integration | Month 2 | Identify approved data sources, develop secure connections, and replace fixed simulated values with authorized planning data |
| Phase 3: Model Improvements | Month 3 | Add manufacturing constraints. Improve knowledge retrieval, and refine AI recommendations |
| Phase 4: Pilot Testing | Month 4 | Test the application with a small group of production planners and collect feedback |
| Phase 5: Performance Evaluation | Month 5 | Measure KPI's, compare results to existing planning methods, and correct identified issues |
| Phase 6: Implementation Review | Month 6 | Review pilot results, complete required security and operational assessments, and determine whether broader implementation is appropriate |

*Table 3. Proposed Six Month Implementation Timeline*

We developed a proposed six month implementation timeline to show how Karate Jones could be expanded beyond the current classroom prototype. This timeline is an estimate for a limited pilot program and would depend on company approval, available resources, and access to the necessary data.

The first month would focus on confirming that the existing workflow performs correctly. This includes testing different demand changes, production disruptions, and inventory levels to make sure the calculations and risk classifications are accurate. During the second and third months, the focus would shift toward improving the application by connecting it to approved data sources and adding more realistic planning constraints.

The fourth month would involve a limited pilot program where a small group of production planners could use the application alongside their existing planning methods. During this phase, the application would remain a decision support tool and would not automatically make changes to production schedules. In the fifth month, the team would evaluate the results using the success factors identified earlier. Finally, the sixth month would focus on reviewing the results and determining whether the application is ready for a larger implementation or requires additional development.

Expanding Karate Jones would require investments in technology, employee time, and system development. However, our team believes these investments could be justified if the application demonstrates measurable improvements in planning efficiency and decision making. One of the main investments would be integrating the application with existing company data systems. This would likely require support from IT employees, database specialists, and production planning personnel. These employees would help identify the correct data sources, establish secure connections, and verify that the information used by the application is accurate. This investment is necessary because the usefulness of the AI recommendations depends heavily on the quality of the data being analyzed. Another investment would involve the AI platform itself. A larger implementation could require additional Dify hosting resources, model usage costs, database storage, and ongoing technical support. As more employees use the application and more scenarios are evaluated, the system would need to handle a greater number of requests while continuing to provide reliable responses. These costs would need to be estimated based on expected usage before a full implementation. Employee training would also be important. Production planners would need to understand how to enter scenarios, interpret risk classifications, and evaluate the recommendations generated by the AI assistant. Training would also need to emphasize that the application supports decisions rather than replacing human judgment. Since the tool would be used in a pharmaceutical manufacturing environment, employees would still be responsible for following established quality, safety, and approval procedures. Finally, ongoing maintenance would be necessary to keep the application useful. Demand patterns, production capabilities, and planning requirements can change over time. The application would need regular updates to its data connections, calculations, knowledge base, and AI instructions. It would also need periodic testing to make sure the recommendations remain accurate and relevant.

Overall, our success factor rubric and scale-up plan provide a way to evaluate whether Karate Jones could be a useful addition to Eli Lilly's production planning process. By measuring calculation accuracy, risk classification accuracy, time savings, recommendation quality, and system reliability, our team can determine whether the application is meeting its intended goals. The scale-up plan also shows that moving from a classroom prototype to an actual business application would require more than simply adding company data. It would involve improving the model, connecting it to existing systems, addressing security requirements, testing it with employees, and making sure the recommendations are reliable. Although our current application

is limited to simulated scenarios, it demonstrates how AI could potentially reduce manual analysis time and help production planners better understand inventory risks. If the application performs well during testing and the expected benefits justify the required investments, it could provide a foundation for a larger production planning decision support system.

**Team Reflection (5 pts)**

Throughout this project, our team learned a lot about how artificial intelligence can be used to improve business processes and support decision making. Before starting the project, we understood that AI could help companies save time and automate certain tasks, but we did not fully understand how much planning and development goes into creating an application that solves a specific business problem. Working with Dify.ai gave us the opportunity to build our own AI application and see how different parts of a workflow can work together to produce useful results. We also learned that AI is not always about replacing tasks that employees perform, but can instead help them better understand information and make more informed decisions.

One of the biggest things our team learned was the importance of clearly defining the problem before developing a solution. Eli Lilly is a large company with many different business processes, so it was challenging to decide which area we wanted to focus on. We initially looked at several challenges related to demand forecasting, manufacturing capacity, and supply chain management. After researching these areas, we decided to focus specifically on production planning for Mounjaro and Zepbound. We felt this was an important area because unexpected changes in demand or production could affect inventory availability, and these changes can be difficult to respond to in the pharmaceutical industry. Narrowing our focus helped us create an application with a more specific purpose instead of trying to solve too many problems at once. Another challenge was developing the Dify.ai workflow itself. Our team had to understand how to connect user inputs, Python calculations, conditional branches, and AI analysis nodes so that the application would function correctly. We learned that each part of the workflow depends on the information passed from the previous step. For example, the Python code calculates projected inventory and determines the risk level, while the IF/ELSE node uses that result to decide which AI analysis node should handle the scenario. This showed us how important it is to organize the workflow correctly and make sure the right information is being passed between nodes. We also learned how specific the instructions given to the AI need to be. Without clear instructions, the AI could provide recommendations that were too general or made assumptions that were not supported by our data. One of the biggest limitations we faced was not having access to Eli Lilly's actual manufacturing and inventory information. Because of this, we had to create our own simulated dataset for Mounjaro and Zepbound. While this allowed us to test the application and demonstrate how it could work, it also meant that our results could not be used to make actual production decisions. We had to be careful to separate our assumptions from real company information and make sure the AI did not present simulated values as facts. This helped us understand how important data accuracy and reliability are when using AI in a business environment, especially in an industry like pharmaceuticals where production decisions can affect patient access to medications.

Our team also learned that creating an AI application involves more than getting the program to produce an answer. We had to consider whether the answer was accurate, understandable, and actually useful to someone making a business decision. This was especially important when developing the different risk categories. We wanted the application to do more than simply tell the user whether inventory was high or low. By creating separate AI analysis nodes for high, medium, and low risk scenarios, we were able to design responses that focus on different planning concerns. We also developed the Karate Jones identity to make the application more engaging while still keeping the recommendations professional and focused on the business problem. If we had more time to continue developing the application, our team would focus on improving the data and making the scenarios more realistic. One improvement would be connecting the application to a database rather than keeping the simulated values directly in the Python code. We would also want to include additional factors such as manufacturing capacity, supplier delays, production lead times, and equipment availability. Another improvement would be connecting our knowledge retrieval results directly to the AI analysis nodes so that the recommendations could use relevant research and approved planning information. Finally, we would want to test the application with people who have experience in production planning to better understand whether the recommendations are practical and what additional information would be useful.

Overall, this project gave our team a better understanding of both the opportunities and limitations of AI in business. We gained experience researching a real company, identifying a specific problem, building an application in Dify.ai, and thinking about how its success could be measured. We also learned the importance of teamwork, problem solving, and being willing to make changes as we better understood the technology. Our biggest takeaway is that AI can be a useful tool for improving business processes, but its effectiveness depends on having reliable data, a clearly defined purpose, and human judgment to evaluate the results. We believe Karate Jones demonstrates how AI could support production planning, and the experience helped us better understand what would be required to turn a classroom project into a tool that could potentially be used in a real business environment.